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The Meaning of "Meaning"

Language is the first broad area of human cognitive capacity for which we are beginning to obtain a description which is not exaggeratedly oversimplified. Thanks to the work of contemporary transformational linguists,[1] a very subtle description of at least some human languages is in the process of being constructed. Some features of these languages appear to be *universal*. Where such features turn out to be "species-specific" — "not explicable on some general grounds of functional utility or simplicity that would apply to arbitrary systems that serve the functions of language" — they may shed some light on the structure of mind. While it is extremely difficult to say to what extent the structure so illuminated will turn out to be a universal structure of *language*, as opposed to a universal structure of innate general learning strategies,[2] the very fact that this discussion can take place is testimony to the richness and generality of the descriptive material that linguists are beginning to provide, and also testimony to the depth of the analysis, insofar as the features that appear to be candidates for "species-specific" features of language are in no sense surface or phenomenological features of language, but lie at the level of deep structure.

The most serious drawback to all of this analysis, as far as a philosopher is concerned, is that it does not concern the meaning of words. Analysis of the deep structure of linguistic forms gives us an incomparably more powerful description of the *syntax* of natural languages than we have ever had before. But the dimension of language associated with the word "meaning" is, in spite of the usual spate of heroic if misguided attempts, as much in the dark as it ever was.

In this essay, I want to explore why this should be so. In my opinion,

[1] The contributors to this area are now too numerous to be listed; the pioneers were, of course, Zellig Harris and Noam Chomsky.

[2] For a discussion of this question, see my "The 'Innateness Hypothesis' and Explanatory Models in Linguistics," *Synthese*, 17(1967): 12–22.

the reason that so-called semantics is in so much worse condition than syntactic theory is that the *prescientific* concept on which semantics is based — the prescientific concept of *meaning* — is itself in much worse shape than the prescientific concept of syntax. As usual in philosophy, skeptical doubts about the concept do not at all help one in clarifying or improving the situation any more than dogmatic assertions by conservative philosophers that all's really well in this best of all possible worlds. The reason that the prescientific concept of meaning is in bad shape is not clarified by some general skeptical or nominalistic argument to the effect that meanings don't exist. Indeed, the upshot of our discussion will be that meanings don't exist in quite the way we tend to think they do. But electrons don't exist in quite the way Bohr thought they did, either. There is all the distance in the world between this assertion and the assertion that meanings (or electrons) "don't exist."

I am going to talk almost entirely about the meaning of words rather than about the meaning of sentences because I feel that our concept of word-meaning is more defective than our concept of sentence-meaning. But I will comment briefly on the arguments of philosophers such as Donald Davidson who insist that the concept of word-meaning *must* be secondary and that study of sentence-meaning must be primary. Since I regard the traditional theories about meaning as myth-eaten (notice that the topic of "meaning" is the one topic discussed in philosophy in which there is literally nothing but "theory" — literally nothing that can be labeled or even ridiculed as the "common sense view"), it will be necessary for me to discuss and try to disentangle a number of topics concerning which the received view is, in my opinion, wrong. The reader will give me the greatest aid in the task of trying to make these matters clear if he will kindly assume that *nothing* is clear in advance.

Meaning and extension. Since the Middle Ages at least, writers on the theory of meaning have purported to discover an ambiguity in the ordinary concept of meaning, and have introduced a pair of terms — *extension* and *intension*, or *Sinn* and *Bedeutung*, or whatever — to disambiguate the notion. The *extension* of a term, in customary logical parlance, is simply the set of things the term is true of. Thus, "rabbit," in its most common English sense, is true of all and only rabbits, so the extension of "rabbit" is precisely the set of rabbits. Even this notion — and it is the *least* problematical notion in this cloudy subject — has its problems, however. Apart from problems it inherits from its parent notion

of *truth*, the forgoing example of "rabbit" *in its most common English sense* illustrates one such problem: strictly speaking, it is not a term, but an ordered pair consisting of a term and a "sense" (or an occasion of use, or something else that distinguishes a term in one sense from the same term used in a different sense) that has an extension. Another problem is this: a "set," in the mathematical sense, is a "yes-no" object; any given object either definitely belongs to S or definitely does not belong to S, if S is a set. But words in a natural language are not generally "yes-no": there are things of which the description "tree" is clearly true and things of which the description "tree" is clearly false, to be sure, but there are a host of borderline cases. Worse, the line between the clear cases and the borderline cases is itself fuzzy. Thus the idealization involved in the notion of *extension* — the idealization involved in supposing that there is such a thing as the set of things of which the term "tree" is true — is actually very severe.

Recently some mathematicians have investigated the notion of a *fuzzy set* — that is, of an object to which other things belong or do not belong with a given probability or to a given degree, rather than belong "yes-no." If one really wanted to formalize the notion of extension as applied to terms in a natural language, it would be necessary to employ "fuzzy sets" or something similar rather than sets in the classical sense.

The problem of a word's having more than one sense is standardly handled by treating each of the senses as a different word (or rather, by treating the word as if it carried invisible subscripts, thus: "rabbit$_1$" — animal of a certain kind; "rabbit$_2$" — coward; and as if "rabbit$_1$" and "rabbit$_2$" or whatever were different words entirely). This again involves two very severe idealizations (at least two, that is): supposing that words have discretely many senses, and supposing that the entire repertoire of senses is fixed once and for all. Paul Ziff has recently investigated the extent to which both of these suppositions distorts the actual situation in natural language;[3] nevertheless, we will continue to make these idealizations here.

Now consider the compound terms "creature with a heart" and "creature with a kidney." Assuming that every creature with a heart possesses a kidney and vice versa, the extension of these two terms is exactly the same. But they obviously differ in meaning. Supposing that there

[3] This is discussed by Ziff in *Understanding Understanding* (Ithaca: Cornell University Press, 1972), especially chap. 1.

is a sense of "meaning" in which meaning = extension, there must be another sense of "meaning" in which the meaning of a term is not its extension but something else, say the "concept" associated with the term. Let us call this "something else" the *intension* of the term. The concept of a creature with a heart is clearly a different concept from the concept of a creature with a kidney. Thus the two terms have different intension. When we say they have different "meaning," meaning = intension.

Intension and extension. Something like the preceding paragraph appears in every standard exposition of the notions "intension" and "extension." But it is not at all satisfactory. Why it is not satisfactory is, in a sense, the burden of this entire essay. But some points can be made at the very outset: first of all, what evidence is there that "extension" *is* a sense of the word "meaning"? The canonical explanation of the notions "intension" and "extension" is very much like "in one sense 'meaning' means *extension* and in the other sense 'meaning' means *meaning*." The fact is that while the notion of "extension" is made quite precise, relative to the fundamental logical notion of *truth* (and under the severe idealizations remarked above), the notion of intension is made no more precise than the vague (and, as we shall see, misleading) notion "concept." It is as if someone explained the notion "probability" by saying: "in one sense 'probability' means frequency, and in the other sense it means *propensity*." "Probability" never means 'frequency', and "propensity" is at least as unclear as "probability."

Unclear as it is, the traditional doctrine that the notion "meaning" possesses the extension/intension ambiguity has certain typical consequences. Most traditional philosophers thought of concepts as something *mental*. Thus the doctrine that the meaning of a term (the meaning "in the sense of intension," that is) is a concept carried the implication that meanings are mental entities. Frege and more recently Carnap and his followers, however, rebelled against this "psychologism," as they termed it. Feeling that meanings are *public* property — that the *same* meaning can be "grasped" by more than one person and by persons at different times — they identified concepts (and hence "intensions" or meanings) with abstract entities rather than mental entities. However, "grasping" these abstract entities was still an individual psychological act. None of these philosophers doubted that understanding a word (knowing its intension) was just a matter of being in a certain

psychological state (somewhat in the way in which knowing how to factor numbers in one's head is just a matter of being in a certain very complex psychological state).

Secondly, the timeworn example of the two terms "creature with a kidney" and "creature with a heart" does show that two terms can have the same extension and yet differ in intension. But it was taken to be obvious that the reverse is impossible: two terms cannot differ in extension and have the same intension. Interestingly, no argument for this impossibility was ever offered. Probably it reflects the tradition of the ancient and medieval philosophers who assumed that the concept corresponding to a term was just a conjunction of predicates, and hence that the concept corresponding to a term must *always* provide a necessary and sufficient condition for falling into the extension of the term.[4] For philosophers like Carnap, who accepted the verifiability theory of meaning, the concept corresponding to a term provided (in the ideal case, where the term had "complete meaning") a *criterion* for belonging to the extension (not just in the sense of "necessary and sufficient condition," but in the strong sense of *way of recognizing* if a given thing falls into the extension or not). Thus these positivistic philosophers were perfectly happy to retain the traditional view on this point. So theory of meaning came to rest on two unchallenged assumptions:

(I) That knowing the meaning of a term is just a matter of being in a certain psychological state (in the sense of "psychological state" in which states of memory and psychological dispositions are "psychological states"; no one thought that knowing the meaning of a word was a

[4] This tradition grew up because the term whose analysis provoked all the discussion in medieval philosophy was the term "God," and the term "God" was thought to be defined through the conjunction of the terms "Good," "Powerful," "Omniscient," etc. — the so-called "Perfections." There was a problem, however, because God was supposed to be a Unity, and Unity was thought to exclude His essence's being complex in any way — i.e., "God" was defined through a conjunction of terms, but God (without quotes) could not be the logical product of properties, nor could He be the unique thing exemplifying the logical product of two or more *distinct* properties, because even this highly abstract kind of "complexity" was held to be incompatible with His perfection of Unity. This is a theological paradox with which Jewish, Arabic, and Christian theologians wrestled for centuries (e.g., the doctrine of the Negation of Privation in Maimonides and Aquinas). It is amusing that theories of contemporary interest, such as conceptualism and nominalism, were first proposed as solutions to the problem of predication in the case of God. It is also amusing that the favorite model of definition in all of this theology — the conjunction-of-properties model — should survive, at least through its consequences, in philosophy of language until the present day.

continuous state of consciousness, of course).

(II) That the meaning of a term (in the sense of "intension") determines its extension (in the sense that sameness of intension entails sameness of extension).

I shall argue that these two assumptions are not jointly satisfied by any notion, let alone any notion of meaning. The traditional concept of meaning is a concept which rests on a false theory.

"Psychological state" and methodological solipsism. In order to show this, we need first to clarify the traditional notion of a psychological state. In one sense a state is simply a two-place predicate whose arguments are an individual and a time. In this sense, *being five feet tall, being in pain, knowing the alphabet,* and even *being a thousand miles from Paris* are all states. (Note that the *time* is usually left implicit or "contextual"; the full form of an atomic sentence of these predicates would be "x is five feet tall at time *t*," "*x is in pain at time t*," etc.) In science, however, it is customary to restrict the term state to properties which are defined in terms of the parameters of the individual which are fundamental from the point of view of the given science. Thus, being five feet tall is a state (from the point of view of physics); being in pain is a state (from the point of view of mentalistic psychology, at least); knowing the alphabet might be a state (from the point of view of cognitive psychology), although it is hard to say; but being a thousand miles from Paris would *not* naturally be called a *state*. In one sense, a psychological state is simply a state which is studied or described by psychology. In this sense it may be trivially true that, say, *knowing the meaning of the word "water"* is a "psychological state" (viewed from the standpoint of cognitive psychology). But this is not the sense of psychological state that is at issue in the above assumption (I).

When traditional philosophers talked about psychological states (or "mental" states), they made an assumption which we may call the assumption of methodological solipsism. This assumption is the assumption that no psychological state, properly so called, presupposes the existence of any individual other than the subject to whom that state is ascribed. (In fact, the assumption was that no psychological state presupposes the existence of the subject's *body* even: if *P* is a psychological state, properly so called, then it must be logically possible for a "disembodied mind" to be in *P*.) This assumption is pretty explicit in Descartes, but it is implicit in just about the whole of traditional philosophical psychology. Making this assumption is, of course, adopting a *restrictive program* — a program which deliberately limits the scope and nature of psychology to fit certain mentalistic preconceptions or, in some cases, to fit an idealistic reconstruction of knowledge and the world. Just *how* restrictive the program is, however, often goes unnoticed. Such common or garden variety psychological states as *being jealous* have to be reconstructed, for example, if the assumption of methodological solipsism is retained. For, in its ordinary use, *x is jealous of* y entails that y exists, and *x is jealous of y's regard for z* entails that both y and z exist (as well as x, of course). Thus *being jealous* and *being jealous of someone's regard for someone else* are not psychological states permitted by the assumption of methodological solipsism. (We shall call them "psychological states in the wide sense" and refer to the states which are permitted by methodological solipsism as "psychological states in the narrow sense.") The reconstruction required by methodological solipsism would be to reconstrue *jealousy* so that I can be jealous of my own hallucinations, or of figments of my imagination, etc. Only if we assume that psychological states in the narrow sense have a significant degree of causal closure (so that restricting ourselves to psychological states in the narrow sense will facilitate the statement of psychological laws) is there any point to engaging in this reconstruction, or in making the assumption of methodological solipsism. But the three centuries of failure of mentalistic psychology is tremendous evidence against this procedure, in my opinion.

Be that as it may, we can now state more precisely what we claimed at the end of the preceding section. Let *A* and *B* be any two terms which differ in extension. By assumption (II) they must differ in meaning (in the sense of "intension"). By assumption (I), *knowing the meaning of A* and *knowing the meaning of B* are psychological states *in the narrow sense* — for this is how we shall construe assumption (I). *But these psychological states must determine the extension of the terms A and B just as much as the meanings ("intensions") do.*

To see this, let us try assuming the opposite. Of course, there cannot be two terms *A* and *B* such that *knowing the meaning of A* is the same state as *knowing the meaning of B* even though *A* and *B* have different extensions. For *knowing the meaning of A* isn't just "grasping the intension" of *A*, whatever that may come to; it is also knowing that the

"intension" that one has "grasped" *is* the intension of A. (Thus, someone who knows the meaning of "wheel" presumably "grasps the intension" of its German synonym "Rad"; but if he doesn't know that the "intension" in question is the intension of *Rad*, he isn't said to "know the meaning of *Rad*.") If A and B are different terms, then *knowing the meaning of A* is a different state from *knowing the meaning of B* whether the meanings of A and B be themselves the same or different. But by the same argument, if I_1 and I_2 are different *intensions* and A is a term, then *knowing that I_1 is the meaning of* A is a different psychological state from *knowing that I_2 is the meaning of* A. Thus, there cannot be two different logically possible worlds L_1 and L_2 such that, say, Oscar is in the *same* psychological state (in the narrow sense) in L_1 and L_2 (in all respects), but in L_1 Oscar understands A as having the meaning I_1 and in L_2 Oscar understands A as having the meaning I_2. (For, if there were, then in L_1 Oscar would be in the psychological state *knowing that I_1 is the meaning of* A and in L_2 Oscar would be in the psychological state *knowing that I_2 is the meaning of* A, and these are different and even — assuming that A has just *one* meaning for Oscar in each world — incompatible psychological states in the narrow sense.)

In short, if S is the sort of psychological state we have been discussing — a psychological state of the form *knowing that I is the meaning of* A, where I is an "intension" and A is a term — then the *same* necessary and sufficient condition for falling into the extension of A "works" in *every* logically possible world in which the speaker is in the psychological state S. For the state S *determines* the intension I, and by assumption (II) the intension amounts to a necessary and sufficient condition for membership in the *extension*.

If our interpretation of the traditional doctrine of intension and extension is fair to Frege and Carnap, then the whole psychologism/Platonism issue appears somewhat a tempest in a teapot, as far as meaning-theory is concerned. (Of course, it is a very important issue as far as general philosophy of mathematics is concerned.) For even if meanings are "Platonic" entities rather than "mental" entities on the Frege-Carnap view, "grasping" those entities is presumably a psychological state (in the narrow sense). Moreover, the psychological state uniquely determines the "Platonic" entity. So whether one takes the "Platonic" entity or the psychological state as the "meaning" would appear to be somewhat a matter of convention. And taking the psychological state to

be the meaning would hardly have the consequence that Frege feared, that meanings would cease to be public. For psychological states are "public" in the sense that different people (and even people in different epochs) can be in the *same* psychological state. Indeed, Frege's argument against psychologism is only an argument against identifying concepts with mental particulars, not with mental entities in general.

The "public" character of psychological states entails, in particular, that if Oscar and Elmer understand a word A differently, then they must be in *different* psychological states. For the state of *knowing the intension of A to be, say I* is the *same* state whether Oscar or Elmer be in it. Thus two speakers cannot be in the same psychological state in all respects and understand the term A differently; the psychological state of the speaker determines the intension (and hence, by assumption (II), the extension) of A.

It is this last consequence of the joint assumptions (I),(II) that we claim to be false. We claim that it is possible for two speakers to be in exactly the *same* psychological state (in the narrow sense), even though the extension of the term A in the idiolect of the one is different from the extension of the term A in the idiolect of the other. Extension is *not* determined by psychological state.

This will be shown in detail in later sections. If this is right, then there are two courses open to one who wants to rescue at least one of the traditional assumptions: to give up the idea that psychological state (in the narrow sense) determines *intension*, or to give up the idea that intension determines extension. We shall consider these alternatives later.

Are meanings in the head? That psychological state does not determine extension will now be shown with the aid of a little science fiction. For the purpose of the following science-fiction examples, we shall suppose that somewhere in the galaxy there is a planet we shall call Twin Earth. Twin Earth is very much like Earth; in fact, people on Twin Earth even speak *English*. In fact, apart from the differences we shall specify in our science-fiction examples, the reader may suppose that Twin Earth is *exactly* like Earth. He may even suppose that he has a *Doppelgänger* — an identical copy — on Twin Earth, if he wishes, although my stories will not depend on this.

Although some of the people on Twin Earth (say, the ones who call themselves "Americans" and the ones who call themselves "Canadians"

and the ones who call themselves "Englishmen," etc.) speak English, there are, not surprisingly, a few tiny differences which we will now describe between the dialects of English spoken on Twin Earth and Standard English. These differences themselves depend on some of the peculiarities of Twin Earth.

One of the peculiarities of Twin Earth is that the liquid called "water" is not H_2O but a different liquid whose chemical formula is very long and complicated. I shall abbreviate this chemical formula simply as XYZ. I shall suppose that XYZ is indistinguishable from water at normal temperatures and pressures. In particular, it tastes like water and it quenches thirst like water. Also, I shall suppose that the oceans and lakes and seas of Twin Earth contain XYZ and not water, that it rains XYZ on Twin Earth and not water, etc.

If a spaceship from Earth ever visits Twin Earth, then the supposition at first will be that "water" has the same meaning on Earth and on Twin Earth. This supposition will be corrected when it is discovered that "water" on Twin Earth is XYZ, and the Earthian spaceship will report somewhat as follows:

"On Twin Earth the word 'water' means XYZ."

(It is this sort of use of the word "means" which accounts for the doctrine that extension is one sense of "meaning," by the way. But note that although "means" does mean something like *has as extension in* this example, one would *not* say

"On Twin Earth the meaning of the word 'water' is XYZ"

unless, possibly, the fact that "water is XYZ" was known to every adult speaker of English on Twin Earth. We can account for this in terms of the theory of meaning we develop below; for the moment we just remark that although the verb "means" sometimes means "has as extension," the nominalization "meaning" never means "extension.")

Symmetrically, if a spaceship from Twin Earth ever visits Earth, then the supposition at first will be that the word "water" has the same meaning on Twin Earth and on Earth. This supposition will be corrected when it is discovered that "water" on Earth is H_2O, and the Twin Earthian spaceship will report:

"On Earth[5] the word 'water' means H_2O."

[5] Or rather, they will report: "On Twin Earth (*the Twin Earthian name for Terra* — H.P.) the word 'water' means H_2O."

Note that there is no problem about the extension of the term "water." The word simply has two different meanings (as we say): in the sense in which it is used on Twin Earth, the sense of water_{TE}, what we call "water" simply isn't water; while in the sense in which it is used on Earth, the sense of water_{E}, what the Twin Earthians call "water" simply isn't water. The extension of "water" in the sense of water_{E} is the set of all wholes consisting of H_2O molecules, or something like that; the extension of water in the sense of water_{TE} is the set of all wholes consisting of XYZ molecules, or something like that.

Now let us roll the time back to about 1750. At that time chemistry was not developed on either Earth or Twin Earth. The typical Earthian speaker of English did not know water consisted of hydrogen and oxygen, and the typical Twin Earthian speaker of English did not know "water" consisted of XYZ. Let Oscar_1 be such a typical Earthian English speaker, and let Oscar_2 be his counterpart on Twin Earth. You may suppose that there is no belief that Oscar_1 had about water that Oscar_2 did not have about "water." If you like, you may even suppose that Oscar_1 and Oscar_2 were exact duplicates in appearance, feelings, thoughts, interior monologue, etc. Yet the extension of the term "water" was just as much H_2O on Earth in 1750 as in 1950; and the extension of the term "water" was just as much XYZ on Twin Earth in 1750 as in 1950. Oscar_1 and Oscar_2 understood the term "water" differently in 1750 *although they were in the same psychological state*, and although, given the state of science at the time, it would have taken their scientific communities about fifty years to discover that they understood the term "water" differently. Thus the extension of the term "water" (and, in fact, its "meaning" in the intuitive preanalytical usage of that term) is *not* a function of the psychological state of the speaker by itself.

But, it might be objected, why should we accept it that the term "water" had the same extension in 1750 and in 1950 (on both Earths)? The logic of natural-kind terms like "water" is a complicated matter, but the following is a sketch of an answer. Suppose I point to a glass of water and say "this liquid is called water" (or "this is called water," if the marker "liquid" is clear from the context). My "ostensive definition" of water has the following empirical presupposition: that the body of liquid I am pointing to bears a certain sameness relation (say, *x is the same liquid as y*, or *x is the* same_{L} *as y*) to most of the stuff I and other speakers in my linguistic community have on other occasions called

"water." If this presupposition is false because, say, I am without knowing it pointing to a glass of gin and not a glass of water, then I do not intend my ostensive definition to be accepted. Thus the ostensive definition conveys what might be called a defeasible necessary and sufficient condition: the necessary and sufficient condition for being water is bearing the relation $same_L$ to the stuff in the glass; but this is the necessary and sufficient condition only if the empirical presupposition is satisfied. If it is not satisfied, then one of a series of, so to speak, "fallback" conditions becomes activated.

The key point is that the relation $same_L$ is a *theoretical* relation: whether something is or is not the same liquid as *this* may take an indeterminate amount of scientific investigation to determine. Moreover, even if a "definite" answer has been obtained either through scientific investigation or through the application of some "common sense" test, the answer is *defeasible*: future investigation might reverse even the most "certain" example. Thus, the fact that an English speaker in 1750 might have called XYZ "water," while he or his successors would not have called XYZ water in 1800 or 1850 does not mean that the "meaning" of "water" changed for the average speaker in the interval. In 1750 or in 1850 or in 1950 one might have pointed to, say, the liquid in Lake Michigan as an example of "water." What changed was that in 1750 we would have mistakenly thought that XYZ bore the relation $same_L$ to the liquid in Lake Michigan, while in 1800 or 1850 we would have known that it did not (I am ignoring the fact that the liquid in Lake Michigan was only dubiously water in 1950, of course).

Let us now modify our science-fiction story. I do not know whether one can make pots and pans out of molybdenum; and if one can make them out of molybdenum, I don't know whether they could be distinguished easily from aluminum pots and pans. (I don't know any of this even though I have acquired the word "molybdenum.") So I shall suppose that molybdenum pots and pans *can't* be distinguished from aluminum pots and pans save by an expert. (To emphasize the point, I repeat that this could be true for all I know, and a fortiori it could be true for all I know by virtue of "knowing the meaning" of the words *aluminum and molybdenum.*) We will now suppose that molybdenum is as common on Twin Earth as aluminum is on Earth, and that aluminum is as rare on Twin Earth as molybdenum is on Earth. In particular, we shall assume that "aluminum" pots and pans are made of molybdenum on Twin Earth. Finally, we shall assume that the words "aluminum" and "molybdenum" are *switched* on Twin Earth: "aluminum" is the name of *molybdenum* and "molybdenum" is the name of *aluminum.*

This example shares some features with the previous one. If a spaceship from Earth visited Twin Earth, the visitors from Earth probably would not suspect that the "aluminum" pots and pans on Twin Earth were not made of aluminum, especially when the Twin Earthians *said* they were. But there is one important difference between the two cases. An Earthian metallurgist could tell very easily that "aluminum" was molybdenum, and a Twin Earthian metallurgist could tell equally easily that aluminum was "molybdenum." (The shudder quotes in the preceding sentence indicate Twin Earthian usages.) Whereas in 1750 no one on either Earth or Twin Earth could have distinguished water from "water," the confusion of aluminum with "aluminum" involves only a part of the linguistic communities involved.

The example makes the same point as the preceding one. If $Oscar_1$ and $Oscar_2$ are standard speakers of Earthian English and Twin Earthian English respectively, and neither is chemically or metallurgically sophisticated, then there may be no difference at all in their psychological state when they use the word "aluminum"; nevertheless we have to say that "aluminum" has the extension *aluminum* in the idiolect of $Oscar_1$ and the extension *molybdenum* in the idiolect of $Oscar_2$. (Also we have to say that $Oscar_1$ and $Oscar_2$ mean different things by "aluminum," that "aluminum" has a different meaning on Earth than it does on Twin Earth, etc.) Again we see that the psychological state of the speaker does *not* determine the extension (or the "meaning," speaking preanalytically) of the word.

Before discussing this example further, let me introduce a *non*-science-fiction example. Suppose you are like me and cannot tell an elm from a beech tree. We still say that the extension of "elm" in my idiolect is the same as the extension of "elm" in anyone else's, viz., the set of all elm trees, and that the set of all beech trees is the extension of "beech" in *both* of our idiolects. Thus "elm" in my idiolect has a different extension from "beech" in your idiolect (as it should). Is it really credible that this difference in extension is brought about by some difference in our *concepts*? My *concept* of an elm tree is exactly the same as my concept of a beech tree (I blush to confess). (This shows that the identifi-

cation of meaning "in the sense of intension" with *concept* cannot be correct, by the way.) If someone heroically attempts to maintain that the difference between the extension of "elm" and the extension of "beech" in *my* idiolect is explained by a difference in my psychological state, then we can always refute him by constructing a "Twin Earth" example — just let the words "elm" and "beech" be switched on Twin Earth (the way "aluminum" and "molybdenum" were in the previous example). Moreover, suppose I have a *Doppelgänger* on Twin Earth who is molecule for molecule "identical" with me (in the sense in which two neckties can be "identical"). If you are a dualist, then also suppose my *Doppelgänger* thinks the same verbalized thoughts I do, has the same sense data, the same dispositions, etc. It is absurd to think *his* psychological state is one bit different from mine: yet he "means" *beech* when he says "elm" and *I* "mean" *elm* when I say elm. Cut the pie any way you like, "meanings" just ain't in the *head*!

A socio-linguistic hypothesis. The last two examples depend upon a fact about language that seems, surprisingly, never to have been pointed out: that there is *division of linguistic labor*. We could hardly use such words as "elm" and "aluminum" if no one possessed a way of recognizing elm trees and aluminum metal; but not everyone to whom the distinction is important has to be able to make the distinction. Let us shift the example: consider *gold*. Gold is important for many reasons: it is a precious metal, it is a monetary metal, it has symbolic value (it is important to most people that the "gold" wedding ring they wear *really* consist of gold and not just *look* gold), etc. Consider our community as a "factory": in this "factory" some people have the "job" of *wearing gold wedding rings*, other people have the "job" of selling gold wedding rings, still other people have the job of *telling whether or not something is really gold*. It is not at all necessary or efficient that everyone who wears a gold ring (or a gold cuff link, etc.), or discusses the "gold standard," etc., engage in buying and selling gold. Nor is it necessary or efficient that everyone who buys and sells gold be able to tell whether or not something is really gold in a society where this form of dishonesty is uncommon (selling fake gold) and in which one can easily consult an expert in case of doubt. And it is *certainly* not necessary or efficient that everyone who has occasion to buy or wear gold be able to tell with any reliability whether or not something is really gold.

The foregoing facts are just examples of mundane division of labor

(in a wide sense). But they engender a division of linguistic labor: everyone to whom gold is important for any reason has to *acquire* the word "gold"; but he does not have to acquire the *method of recognizing* if something is or is not gold. He can rely on a special subclass of speakers. The features that are generally thought to be present in connection with a general name — necessary and sufficient conditions for membership in the extension, ways of recognizing if something is in the extension ("criteria"), etc. — are all present in the linguistic community *considered as a collective body*; but that collective body divides the "labor" of knowing and employing these various parts of the "meaning" of "gold."

This division of linguistic labor rests upon and presupposes the division of *non*-linguistic labor, of course. If only the people who know how to tell if some metal is really gold or not have any reason to have the word "gold" in their vocabulary, then the word "gold" will be as the word "water" was in 1750 with respect to that subclass of speakers, and the other speakers just won't acquire it at all. And some words do not exhibit any division of linguistic labor: "chair," for example. But with the increase of division of labor in the society and the rise of science, more and more words begin to exhibit this kind of division of labor. "Water," for example, did not exhibit it at all prior to the rise of chemistry. Today it is obviously necessary for every speaker to be able to recognize water (reliably under normal conditions), and probably every adult speaker even knows the necessary and sufficient condition "water is H_2O," but only a few adult speakers could distinguish water from liquids which superficially resembled water. In case of doubt, other speakers would rely on the judgment of these "expert" speakers. Thus the way of recognizing possessed by these "expert" speakers is also, through them, possessed by the collective linguistic body, even though it is not possessed by each individual member of the body, and in this way the most recherché fact about water may become part of the *social* meaning of the word while being unknown to almost all speakers who acquire the word.

It seems to me that this phenomenon of division of linguistic labor is one which it will be very important for sociolinguistics to investigate. In connection with it, I should like to propose the following hypothesis:

HYPOTHESIS OF THE UNIVERSALITY OF THE DIVISION OF LINGUISTIC LABOR:

> Every linguistic community exemplifies the sort of division of linguistic labor just described, that is, possesses at least some terms whose associated "criteria" are known only to a subset of the speakers who acquire the terms, and whose use by the other speakers depends upon a structured cooperation between them and the speakers in the relevant subsets.

It would be of interest, in particular, to discover if extremely primitive peoples were sometimes exceptions to this hypothesis (which would indicate that the division of linguistic labor is a product of social evolution), or if even they exhibit it. In the latter case, one might conjecture that division of labor, including linguistic labor, is a fundamental trait of our species.

It is easy to see how this phenomenon accounts for some of the examples given above of the failure of the assumptions (I), (II). Whenever a term is subject to the division of linguistic labor, the "average" speaker who acquires it does not acquire anything that fixes its extension. In particular, his individual psychological state *certainly* does not fix its extension; it is only the sociolinguistic state of the collective linguistic body to which the speaker belongs that fixes the extension.

We may summarize this discussion by pointing out that there are two sorts of tools in the world: there are tools like a hammer or a screwdriver which can be used by one person; and there are tools like a steamship which require the cooperative activity of a number of persons to use. Words have been thought of too much on the model of the first sort of tool.

Indexicality and rigidity.[6] The first of our science-fiction examples — "water" on Earth and on Twin Earth in 1750 — does not involve division of linguistic labor, or at least does not involve it in the same way the examples of "aluminum" and "elm" do. There were not (in our story, anyway) any "experts" on water on Earth in 1750, nor any experts on "water" on Twin Earth. (The example *can* be construed as involving division of labor *across time*, however. I shall not develop this method of treating the example here.) The example *does* involve

[6] The substance of this section was presented at a series of lectures I gave at the University of Washington (Summer Institute in Philosophy) in 1968, and at a lecture at the University of Minnesota (at the conference out of which this volume originated).

things which are of fundamental importance to the theory of reference and also to the theory of necessary truth, which we shall now discuss.

There are two obvious ways of telling someone what one means by a natural-kind term such as "water" or "tiger" or "lemon." One can give him a so-called ostensive definition — "this (liquid) is water"; "this (animal) is a tiger"; "this (fruit) is a lemon"; where the parentheses are meant to indicate that the "markers" *liquid, animal, fruit,* may be either explicit or implicit. Or one can give him a *description.* In the latter case the description one gives typically consists of one or more markers together with a *stereotype*[7] — a standardized description of features of the kind that are typical, or "normal," or at any rate stereotypical. The central features of the stereotype generally are *criteria* — features which in normal situations constitute ways of recognizing if a thing belongs to the kind or, at least, necessary conditions (or probabilistic necessary conditions) for membership in the kind. Not all criteria used by the linguistic community as a collective body are included in the stereotype, and in some cases the stereotype may be quite weak. Thus (unless I am a very atypical speaker), the stereotype of an elm is just that of a common deciduous tree. These features are indeed necessary conditions for membership in the kind (I mean "necessary" in a loose sense; I don't think "elm trees are deciduous" is *analytic*), but they fall far short of constituting a way of recognizing elms. On the other hand, the stereotype of a tiger does enable one to recognize tigers (unless they are albino, or some other atypical circumstance is present), and the stereotype of a lemon generally enables one to recognize lemons. In the extreme case, the stereotype may be *just* the marker: the stereotype of molybdenum might be *just* that molybdenum is a *metal.* Let us consider both of these ways of introducing a term into someone's vocabulary.

Suppose I point to a glass of liquid and say "*this* is water," in order to teach someone the word "water." We have already described some of the empirical presuppositions of this act, and the way in which this kind of meaning-explanation is defeasible. Let us now try to clarify further how it is supposed to be taken.

In what follows, we shall take the notion of "possible world" as primitive. We do this because we feel that in several senses the notion makes

[7] See my "Is Semantics Possible," *Metaphilosophy,* 1, no. 3 (July 1970).

sense and is scientifically important even if it needs to be made more precise. We shall assume further that in at least some cases it is possible to speak of the same individual as existing in more than one possible world.[8] Our discussion leans heavily on the work of Saul Kripke, although the conclusions were obtained independently.

Let W_1 and W_2 be two possible worlds in which I exist and in which this glass exists and in which I am giving a meaning explanation by pointing to this glass and saying "this is water." (We do *not* assume that the *liquid* in the glass is the same in both worlds.) Let us suppose that in W_1 the glass is full of H_2O and in W_2 the glass is full of XYZ. We shall also suppose that W_1 is the *actual* world and that XYZ is the stuff typically called "water" in the world W_2 (so that the relation between English speakers in W_1 and English speakers in W_2 is exactly the same as the relation between English speakers on Earth and English speakers on Twin Earth). Then there are two theories one might have concerning the meaning of "water."

(1) One might hold that "water" was *world-relative* but *constant* in meaning (i.e., the word has a *constant relative meaning*). On this theory, "water" *means the same* in W_1 and W_2; it's just that water is H_2O in W_1 and water is XYZ in W_2.

(2) One might hold that water is H_2O in all worlds (the stuff called "water" in W_2 isn't water), but "water" doesn't have the same meaning in W_1 and W_2.

If what was said before about the Twin Earth case was correct, then (2) is clearly the correct theory. When I say "*this* (liquid) is water," the "this" is, so to speak, a *de re* "this" — i.e., the force of my explanation is that "water" is whatever bears a certain equivalence relation (the relation we called "same$_L$" above) to the piece of liquid referred to as "this" *in the actual world.*

We might symbolize the difference between the two theories as a "scope" difference in the following way. On theory (1), the following is true:

(1′) (For every world W) (For every x in W) (x is water $\equiv$ x bears same$_L$ to the entity referred to as "this" in W)

while on theory (2):

[8] This assumption is not actually needed in what follows. What is needed is that the same *natural kind* can exist in more than one possible world.

(2′) (For every world W) (For every x in W) (x is water $\equiv$ x bears same$_L$ to the entity referred to as "this" *in the actual world* W_1).

(I call this a "scope" difference because in (1′) "the entity referred to as 'this' " is within the scope of "For every world W" — as the qualifying phrase "in W" makes explicit, whereas in (2′) "the entity referred to as 'this' " means "the entity referred to as 'this' *in the actual world,*" and has thus a reference *independent* of the bound variable "W.")

Kripke calls a designator "rigid" (in a given sentence) if (in that sentence) it refers to the same individual in every possible world in which the designator designates. If we extend the notion of rigidity to substance names, then we may express Kripke's theory and mine by saying that the term "water" is *rigid.*

The rigidity of the term "water" follows from the fact that when I give the ostensive definition "*this* (liquid) is water" I intend (2′) and not (1′).

We may also say, following Kripke, that when I give the ostensive definition "*this* (liquid) is water," the demonstrative "this" is *rigid.*

What Kripke was the first to observe is that this theory of the meaning (or "use," or whatever) of the word "water" (and other natural-kind terms as well) has startling consequences for the theory of necessary truth.

To explain this, let me introduce the notion of a *cross-world relation.* A two-term relation R will be called *cross-world* when it is understood in such a way that its extension is a set of ordered pairs of individuals *not all in the same possible world.* For example, it is easy to understand the relation *same height as* as a cross-world relation: just understand it so that, e.g., if x is an individual in a world W_1 who is five feet tall (in W_1) and y is an individual in W_2 who is five feet tall (in W_2), then the ordered pair x,y belongs to the extension of *same height as.* (Since an individual may have different heights in different possible worlds in which that same individual exists, strictly speaking it is not the ordered pair x,y that constitutes an element of the extension of *same height as,* but rather the ordered pair *x-in-world-*W_1, *y-in-world-*W_2.)

Similarly, we can understand the relation *same$_L$* (same liquid as) as a cross-world relation by understanding it so that a liquid in world W_1

which has the same important physical properties (in W_1) that a liquid in W_2 possesses (in W_2) bears $same_L$ to the latter liquid.

Then the theory we have been presenting may be summarized by saying that an entity x, in an arbitrary possible world, is *water* if and only if it bears the relation $same_L$ (construed as a cross-world relation) to the stuff we call "water" in the *actual* world.

Suppose, now, that I have not yet discovered what the important physical properties of water are (in the actual world) — i.e., I don't yet know that water is H_2O. I may have ways of *recognizing* water that are successful (of course, I may make a small number of mistakes that I won't be able to detect until a later stage in our scientific development) but not know the microstructure of water. If I agree that a liquid with the superficial properties of "water" but a different microstructure *isn't really water*, then my ways of recognizing water (my "operational definition," so to speak) cannot be regarded as an analytical specification of *what it is to be* water. Rather, the operational definition, like the ostensive one, is simply a way of pointing out a standard — pointing out the stuff *in the actual world* such that for x to be water, in *any* world, is for x to bear the relation $same_L$ to the *normal* members of the class of *local* entities that satisfy the operational definition. "Water" on Twin Earth is not water, even if it satisfies the operational definition, because it doesn't bear $same_L$ to the *local* stuff that satisfies the operational definition, and local stuff that satisfies the operational definition but has a microstructure different from rest of the local stuff that satisfies the operational definition isn't water either, because it doesn't bear $same_L$ to the *normal* examples of the local "water."

Suppose, now, that I discover the microstructure of water — that water is H_2O. At this point I will be able to say that the stuff on Twin Earth that I earlier *mistook* for water isn't really water. In the same way, if you describe not another planet in the actual universe, but another possible universe in which there is stuff with the chemical formula XYZ which passes the "operational test" for *water*, we shall have to say that that stuff isn't water but merely XYZ. You will not have described a possible world in which "water is XYZ," but merely a possible world in which there are lakes of XYZ, people drink XYZ (and not water), or whatever. In fact, once we have discovered the nature of water, nothing counts as a possible world in which water doesn't have that nature. Once we have discovered that water (in the actual world) is

H_2O, *nothing counts as a possible world in which water isn't* H_2O. In particular, if a "logically possible" statement is one that holds in some "logically possible world," *it isn't logically possible that water isn't* H_2O.

On the other hand, we can perfectly well imagine having experiences that would convince us (and that would make it rational to believe that) water *isn't* H_2O. In that sense, it is conceivable that water isn't H_2O. It is conceivable but it isn't logically possible! Conceivability is no proof of logical possibility.

Kripke refers to statements which are rationally unrevisable (assuming there are such) as *epistemically necessary*. Statements which are true in all possible worlds he refers to simply as necessary (or sometimes as "metaphysically necessary"). In this terminology, the point just made can be restated as: a statement can be (metaphysically) necessary and epistemically contingent. Human intuition has no privileged access to metaphysical necessity.

Since Kant there has been a big split between philosophers who thought that all necessary truths were analytic and philosophers who thought that some necessary truths were synthetic a priori. But none of these philosophers thought that a (metaphysically) necessary truth could fail to be a priori: the Kantian tradition was as guilty as the empiricist tradition of equating metaphysical and epistemic necessity. In this sense Kripke's challenge to received doctrine goes far beyond the usual empiricism/Kantianism oscillation.

In this paper our interest is in theory of meaning, however, and not in theory of necessary truth. Points closely related to Kripke's have been made in terms of the notion of *indexicality*.[9] Words like "now," "this," "here," have long been recognized to be *indexical*, or *token-reflexive* — i.e., to have an extension which varied from context to context or token to token. For these words no one has ever suggested the traditional theory that "intension determines extension." To take our Twin Earth example: if I have a *Doppelgänger* on Twin Earth, then when I think "I have a headache," *he* thinks "I have a headache." But the extension of the particular token of "I" in his verbalized thought is himself (or his unit class, to be precise), while the extension of the token of "I" in *my* verbalized thought is *me* (or my unit class, to be precise). So the same word, "I," has two different extensions in two different idio-

[9] These points were made in my 1968 lectures at the University of Washington and the University of Minnesota.